

mse测试结果

代码部分

long/short, rebalance设计

```
def compute_portfolio_returns(  
    preds: np.ndarray, targets: np.ndarray  
    ) -> Dict[str, float]:  
    """  
    Rebalance & Return 逻辑:  
    - 每月按预测值排序, Top 10% 做多, Bottom 10% 做空。  
    - 组合收益 = long mean - short mean.  
    """  
    n = preds.size # preds: 预测值  
    if n == 0:  
        return {"long": np.nan, "short": np.nan, "long_short": np.nan}  
    order = np.argsort(preds)  
    k = max(1, int(np.floor(n * 0.1)))  
    long_idx = order[-k:]  
    short_idx = order[:k]  
    long_ret = float(np.mean(targets[long_idx])) if long_idx.size else float("nan")  
    short_ret = float(np.mean(targets[short_idx])) if short_idx.size else float("nan")  
    long_short = (  
        float(long_ret - short_ret)  
        if np.isfinite(long_ret) and np.isfinite(short_ret)  
        else float("nan")  
    )  
    return {"long": long_ret, "short": short_ret, "long_short": long_short}
```

训练时间：选择1989.12-1994.12五年数据，仅预测1995.1-1995.6 六个月

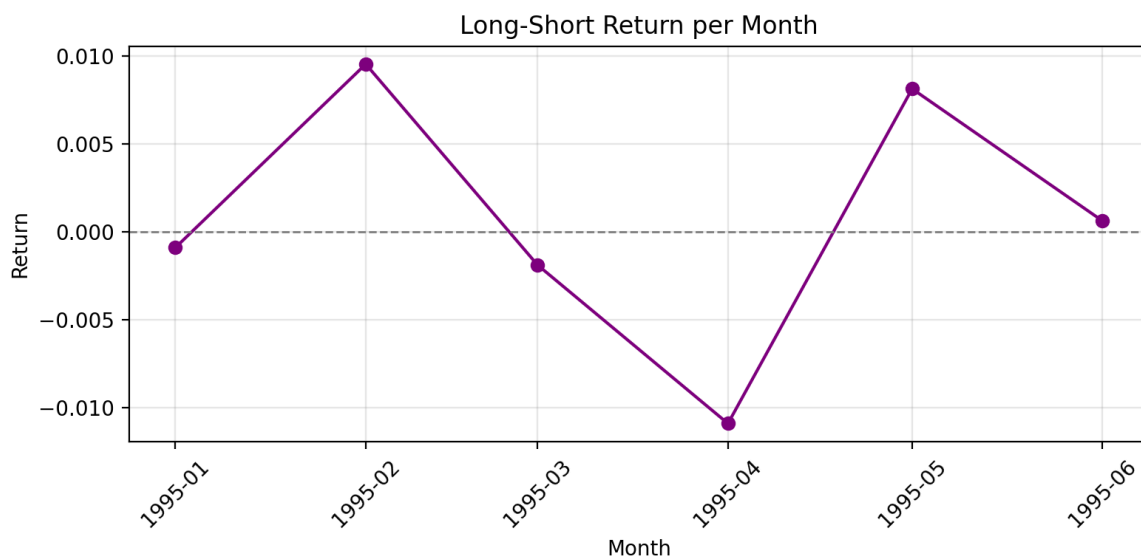
预测结果

Rebalancing monthly: Top 10% predictions → Long, Bottom 10% → Short.
Month 1995-01 | MSE 0.144110 | R2 -2.2853 | Long 0.0372 | Short 0.0381 | Long-Short -0.0009
Month 1995-02 | MSE 0.083519 | R2 -2.5766 | Long 0.0303 | Short 0.0207 | Long-Short 0.0096
Month 1995-03 | MSE 0.253002 | R2 -9.0932 | Long 0.0252 | Short 0.0271 | Long-Short -0.0019
Month 1995-04 | MSE 0.098865 | R2 -2.4386 | Long 0.0263 | Short 0.0372 | Long-Short -0.0109

Month 1995-05 | MSE 0.297344 | R2 -8.4381 | Long 0.0554 | Short 0.0472
| Long-Short 0.0081
Month 1995-06 | MSE 1.174731 | R2 -37.9274 | Long 0.0593 | Short 0.0586 |
Long-Short 0.0006
Saved metrics for MSE to sanity_outputs/sanity_metrics_mse.csv
Wrote summary to sanity_outputs/sanity_summary_mse.json
Loss curve saved to sanity_outputs/mse_loss_curve.png
Long-Short return curve saved to sanity_outputs/mse_returns_curve.png
MSE 实验完成。

画图

return:



MSE 值

